EXPERIMENT-2

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USER INTERFACE AND DESIGN

Develop and compare CLI, GUI, and Voice User Interfaces (VUI) for the same task and assess user satisfaction using Python (Tkinter for GUI, Speech Recognition for VUI), Terminal

AIM:

The aim is to develop and compare Command Line Interface (CLI), Graphical User Interface (GUI), and Voice User Interface (VUI) for the same task, and assess user satisfaction using Python (with Tkinter for GUI and Speech Recognition for VUI) and Terminal.

PROCEDURE:

A movie watchlist system is developed using three interfaces: CLI, GUI, and VUI. In the CLI version, users manage their watchlist through terminal commands such as adding, viewing, and removing movies. In the GUI version, Tkinter is used to create a user-friendly window where users can enter movie names, display the list, and delete selected movies using buttons. In the VUI version, Speech Recognition is used to accept voice commands like “add movie Inception” or “remove movie Avatar” and update the watchlist accordingly. After implementing all three interfaces, users test each version and provide feedback through a satisfaction survey. The collected responses are analyzed to compare usability, speed, and user preference across CLI, GUI, and VUI implementations.

i) CLI (Command Line Interface)

CLI implementation where users can add, view, and remove movies using the terminal.

```
watchlist = []
def add_movie():
    movie = input("Enter movie name: ")
    if movie.strip():
        watchlist.append(movie)
        print(f"'{movie}' added to watchlist.")
    else:
        print("Movie name cannot be empty.")

def view_movies():
    if not watchlist:
        print("Watchlist is empty.")
        return
    print("\n🎬 Your Watchlist:")
    for i, movie in enumerate(watchlist, start=1):
        print(f"{i}. {movie}")

def remove_movie():
    view_movies()
    if not watchlist:
        return
    try:
        num = int(input("Enter movie number to remove: "))
        if 1 <= num <= len(watchlist):
            removed = watchlist.pop(num - 1)
            print(f" Removed '{removed}' from watchlist.")
        else:
```

```
            print("Invalid number.")
    except ValueError:
        print("Please enter a valid number.")

def main():
    while True:
        print("\n===== MOVIE WATCHLIST (CLI) =====")
        print("1. Add Movie")
        print("2. View Watchlist")
        print("3. Remove Movie")
        print("4. Exit")
        choice = input("Choose an option: ")


        if choice == "1":
            add_movie()
        elif choice == "2":
            view_movies()
        elif choice == "3":
            remove_movie()
        elif choice == "4":
            print("Exiting... Enjoy your movies!")
            break
        else:
            print("Invalid choice. Try again.")

if __name__ == "__main__":
    main()
```

Sample Output :

```
===== MOVIE WATCHLIST (CLI) =====
1. Add Movie
2. View Watchlist
3. Remove Movie
4. Exit
Choose an option: 1
Enter movie name: Remo
'Remo' added to watchlist.

===== MOVIE WATCHLIST (CLI) =====
1. Add Movie
2. View Watchlist
3. Remove Movie
4. Exit
Choose an option: 1
Enter movie name: Don
'Don' added to watchlist.

===== MOVIE WATCHLIST (CLI) =====
1. Add Movie
2. View Watchlist
3. Remove Movie
4. Exit
Choose an option: 1
Enter movie name: Doctor
'Doctor' added to watchlist.

===== MOVIE WATCHLIST (CLI) =====
1. Add Movie
2. View Watchlist
3. Remove Movie
4. Exit
Choose an option: 2

🎬 Your Watchlist:
1. Remo
2. Don
3. Doctor
```

```
===== MOVIE WATCHLIST (CLI) =====
1. Add Movie
2. View Watchlist
3. Remove Movie
4. Exit
Choose an option: 3

🎬 Your Watchlist:
1. Remo
2. Don
3. Doctor
Enter movie number to remove: 2
🗑️ Removed 'Don' from watchlist.

===== MOVIE WATCHLIST (CLI) =====
1. Add Movie
2. View Watchlist
3. Remove Movie
4. Exit
Choose an option: 2

🎬 Your Watchlist:
1. Remo
2. Doctor

===== MOVIE WATCHLIST (CLI) =====
1. Add Movie
2. View Watchlist
3. Remove Movie
4. Exit
Choose an option: 4
Exiting... Enjoy your movies!
```

ii) GUI (Graphical User Interface)

Tkinter to create a simple GUI for our Movie Watchlist application.

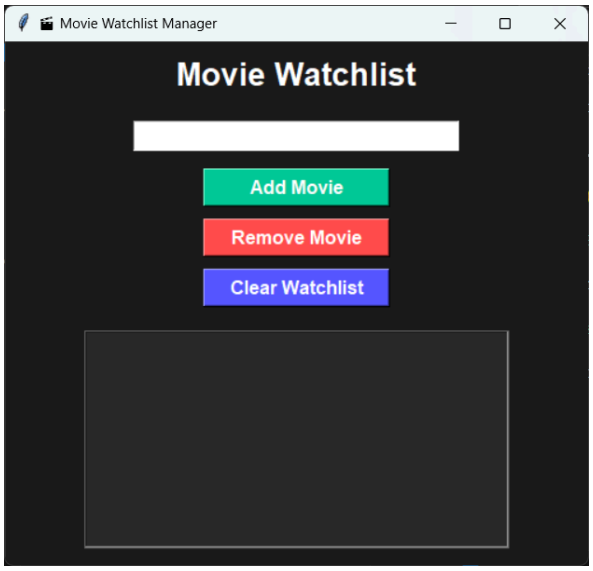
```
import tkinter as tk
from tkinter import messagebox

watchlist = []
def update_count():
    count_label.config(text=f"Total Movies:
{len(watchlist)}")

def add_movie():
    movie = movie_entry.get()
    if movie.strip() == "":
        messagebox.showwarning("Warning", "Movie name cannot
be empty!")
        return
    watchlist.append(movie)
    listbox.insert(tk.END, movie)
    movie_entry.delete(0, tk.END)
    update_count()

def remove_movie():
    selected = listbox.curselection()
    if not selected:
        messagebox.showinfo("Info", "Select a movie to
remove.")
        return
    index = selected[0]
    removed_movie = watchlist.pop(index)
    listbox.delete(index)
    messagebox.showinfo("Removed", f"'{removed_movie}'
```

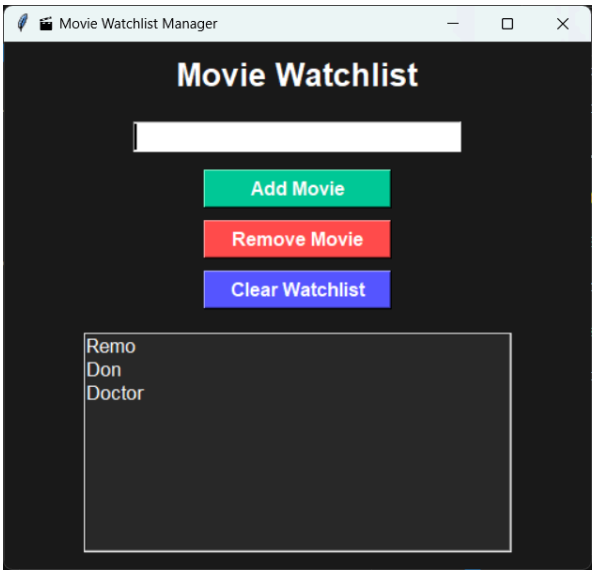
```
removed from watchlist!")
    update_count()

def clear_all():
    if not watchlist:
        messagebox.showinfo("Info", "Watchlist is already
empty.")
        return
    confirm = messagebox.askyesno("Confirm", "Do you want to
clear the entire watchlist?")
    if confirm:
        watchlist.clear()
        listbox.delete(0, tk.END)
        update_count()

# Window Setup
root = tk.Tk()
root.title("Movie Watchlist Manager")
root.geometry("520x520")
root.configure(bg="#121212")

# Title
title = tk.Label(root, text="Movie Watchlist", font=("Arial",
22, "bold"),fg="white", bg="#121212")
title.pack(pady=15)

# Entry
movie_entry = tk.Entry(root, font=("Arial", 14), width=28,
bg="#2c2c2c",fg="white", insertbackground="white")
movie_entry.pack(pady=10)
```

```
# Buttons Frame
btn_frame = tk.Frame(root, bg="#121212")
btn_frame.pack(pady=10)


add_btn = tk.Button(btn_frame, text="Add Movie",
font=("Arial", 12, "bold"),bg="#00c896", fg="white",
width=12, command=add_movie)
add_btn.grid(row=0, column=0, padx=10)



remove_btn = tk.Button(btn_frame, text="Remove Movie",
font=("Arial", 12, "bold"),bg="#ff4d4d", fg="white",
width=12,command=remove_movie)
remove_btn.grid(row=0, column=1, padx=10)


clear_btn = tk.Button(btn_frame, text="Clear Watchlist",
font=("Arial", 12, "bold"),bg="#5555ff", fg="white",
width=12, command=clear_all)
clear_btn.grid(row=0, column=2, padx=10)


listbox = tk.Listbox(root, font=("Arial", 13), width=40,
height=12,bg="#1f1f1f",
fg="white",selectbackground="#ffcc00",
selectforeground="black")
listbox.pack(pady=20)


count_label = tk.Label(root, text="Total Movies: 0",
font=("Arial", 14, "bold")fg="white", bg="#121212")
count_label.pack(pady=10)
root.mainloop()
```

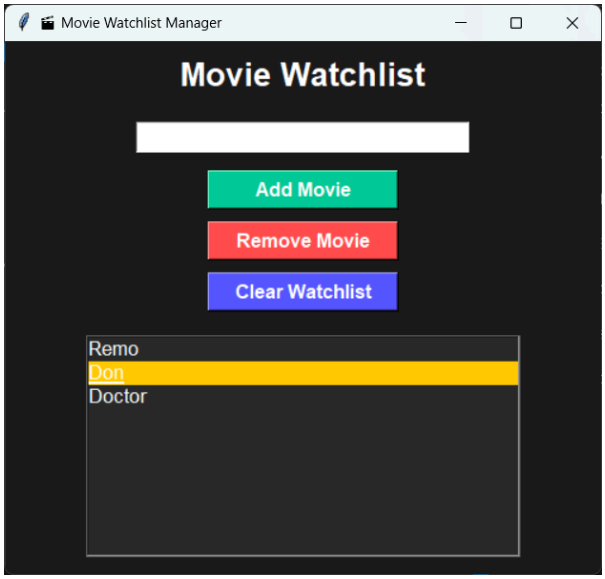
Sample Output :

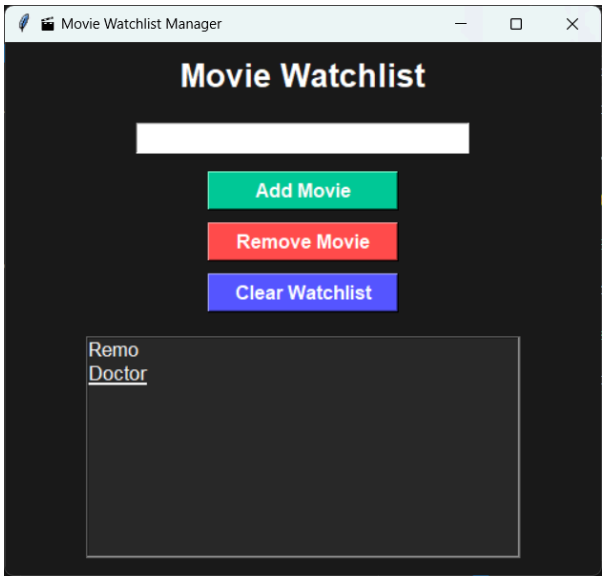
ADD MOVIE





REMOVE MOVIE





iii) VUI (Voice User Interface)

speech_recognition library for voice input and the pyttsx3 library for text-to-speech output. Make sure you have these libraries installed (pip install SpeechRecognition pyttsx3).

```
import speech_recognition as sr
import pyttsx3

watchlist = []

engine = pyttsx3.init()
recognizer = sr.Recognizer()

def speak(text):
    print("Assistant:", text)
    engine.say(text)
    engine.runAndWait()

def listen():
    with sr.Microphone(device_index=21) as source:
        print("Listening...")
        recognizer.adjust_for_ambient_noise(source)
        try:
                audio = recognizer.listen(source, timeout=10,
phrase_time_limit=8)
        except sr.WaitTimeoutError:
            print("No voice detected")
            return ""
    try:
        command = recognizer.recognize_google(audio)
```

```
        print("You said:", command)
        return command.lower()
    except sr.UnknownValueError:
        speak("Sorry, I did not understand.")
        return ""
    except sr.RequestError:
        speak("Network error. Please check internet connection.")
        return ""

def add_movie():
    speak("Tell me the movie name.")
    movie = listen()
    if movie.strip():
        watchlist.append(movie)
        speak(f"{movie} added to watchlist.")
    else:
        speak("Movie name cannot be empty.")

def view_movies():
    if not watchlist:
        speak("Your watchlist is empty.")
        return
    speak("Your watchlist contains:")
    for movie in watchlist:
        speak(movie)

def remove_movie():
    if not watchlist:
        speak("Watchlist is empty.")
        return
    speak("Tell the movie name to remove.")
```

```
    movie_name = listen()
    for movie in watchlist:
        if movie_name in movie.lower():
            watchlist.remove(movie)
            speak(f"{movie} removed from watchlist.")
            return
    speak("Movie not found.")

def main():
    speak("Welcome")
    while True:
        speak("Say add, view, remove, or exit.")
        command = listen()
        if command == "":
            continue
        if "add" in command:
            add_movie()
        elif "view" in command:
            view_movies()
        elif "remove" in command:
            remove_movie()
        elif "exit" in command:
            speak("Goodbye")
            break
        else:
    speak("Invalid command. Please say add, view, remove, or
exit.")

if __name__ == "__main__":
    main()
```

Sample Output :

ADD MOVIE

```
Assistant: Welcome
Assistant: Say add, view, remove, or exit.
Listening...
Assistant: Sorry, I did not understand.
Assistant: Say add, view, remove, or exit.
Listening...
You said: add movie
Assistant: Tell me the movie name.
Listening...
You said: Robo
Assistant: robo added to watchlist.
```

```
Assistant: Say add, view, remove, or exit.
Listening...
You said: add a movie
Assistant: Tell me the movie name.
Listening...
You said: Enthiran
Assistant: enthiran added to watchlist.
Assistant: Say add, view, remove, or exit.
Listening...
You said: add movie
Assistant: Tell me the movie name.
Listening...
You said: Robo
Assistant: robo added to watchlist.
```

REMOVE MOVIE

```
Assistant: Say add, view, remove, or exit.
Listening...
You said: remove movie
Assistant: Tell the movie name to remove.
Listening...
You said: Robo
Assistant: robo removed from watchlist.
```

VIEW MOVIE

```
Assistant: Say add, view, remove, or exit.
Listening...
You said: view movie
Assistant: Your watchlist contains:
Assistant: enthiran
Assistant: robo
```

ADD, VIEW and EXIT

```
Assistant: Say add, view, remove, or exit.
Listening...
You said: add movie
Assistant: Tell me the movie name.
Listening...
You said: Remo
Assistant: remo added to watchlist.
```

```
Assistant: Say add, view, remove, or exit.
Listening...
You said: view movie
Assistant: Your watchlist contains:
Assistant: enthiran
Assistant: robo
Assistant: remo
Assistant: Say add, view, remove, or exit.
Listening...
You said: exit
Assistant: Goodbye
```